ASSIGNMENT

1 Find the smallest multiple of 10 which has remainder 2 when divided by 3, and remainder 3 when divided by 7.

Solution:

Form first condition; $n = 10k$

Form 2nd condition; $n \equiv 2 \bmod (3)$

$10k \equiv 2 \bmod (3)$

$k \equiv 2 \bmod (3)$

min. $k = 2$

$\Rightarrow n = 20 + 30\,m$

From 3rd condition $n \equiv 3 \bmod (7)$

$20 + 30m \equiv 3 \bmod (7)$

$6 + 2m \equiv 3 \bmod (7)$

$2m \equiv 4 \bmod (7)$

$m \equiv 2 \bmod (7)$

Min. $m = 2$

$\Rightarrow n = 20 + 30 \times 2 + 210\,r$

$n = 80 + 210r$

min. $n = 80$

2 (CHINA/2002) When a positive integer n is divided by 5,7,9,11,the remainders are 1,2,3,4 respectively. Find the minimum value of n.

Solution:

Let n be the solution $n \equiv 1 \pmod 5$ implies $n = 5k + 1$ for some non-negative integer k ;

Then $n \equiv 2 \pmod 7$ implies $5k \equiv 1 \pmod 7$, so minimum k is 3, i.e. $n = 16$ is the minimum n satisfying the first two equations.

Then $n = 16 + 35m$ is the general form of n.

The requirement $n \equiv 3 \pmod 9$

$\Rightarrow 16 + 35m \equiv 3 \pmod 9$ and

Its minimum solution is $m = 4$, so $n = 156 + 315p$, for some p

Finally, from the fourth requirement $n \equiv 4$ (mood11), we obtain $2 + 315\,p \equiv 4 \pmod{11}$, so $7p \equiv 2 \pmod{11}$, i.e.

p = 5 Thus,

N = 156 + 315 × 5 = 1731

3 (CHINA/2003)) Find the integer solutions of the equation 6xy + 4x – 9y – 7 = 0.

Solution:

By factorization, 6xy + 4x – 9y – 6 = (2x – 3)(3y + 2), so the given equation becomes (2x – 3)(3y + 2) = 1.

If 2x – 3 = 1 & 3y + 2 = 1, then y has no integer solution.

If 2x – 3 = –1& 3y + 2 = –1, then x = 1, y = –1.

By checking, (1,–1)satisfies the original equation, so it is the unique solution for (x; y).

4 (CHNMOL/2005) p, q are two integers, and the two roots of the equation in

$x^2 - \frac{p^2+11}{9}x + \frac{15}{4}(p+q) + 16 = 0$ are p and q also. Find the values of p and q.

Solution:

Viete Theorem yields

$p+q = \frac{p^2+11}{9}$......(i)

$pq = \frac{15}{4}(p+q) + 16$......(ii)

Then p + q > 0 and pq> 0 from given equations, so p, q are both positive integers.

From (ii)

16pq – 60(p + q) = 16^2

∴ (4p – 15)(4q – 15) = 256 + 225 = 481

Since 481 = 1 × 481 = 13 × 37 = (–1) × (–481) = (–13) × (–37), and 4p – 15 or 4q – 15 cannot be –37 or –481, so the pair (4p – 15; 4q – 15) has the following four possible cases:

(1; 481); (481; 1); (13; 37); (37; 37).

Corresponding to them, the pairs of (p; q) are(4; 124); (124; 4); (7; 13); (13; 7):

By checking, only the pair (13; 7) satisfies the original system: the equation becomes $x^2 - 20x + 91 = 0$ and its roots are (13; 7). Thus, the solution for (p; q) is (13; 7).

5 (KIEV/1962) Prove that the equation $x^2 + y^2 = 3z^2$has no integer solution (x; y; z) ≠ (0; 0; 0).

Solution:

First of all it can be shown that, if an integer solution (x; y; z) is not (0; 0; 0), then there must be such an integer solution with (x; y) = 1.

Suppose that (x; y; z) ≠ (0; 0; 0) is an integer solution with (x; y) = d > 1, letting

$x = dx_1,\ y = dy_1,\ \text{with}\ (x_1, y_1) = 1$.

Then the original equation becomes $d^2\left(x_1^2+y_1^2\right)=3z^2$, so $d^2 \mid 3z^2$. Since the indices of 3 in d^2 and z^2 are both even so $(d^2,3) = 1$ and $d^2 \mid z^2$, i.e. $d \mid z$. Let $z = dz_1$, then $x_1^2+y_1^2=3z_1^2$. Thus $(x_1, y_1, z_1) \neq (0; 0; 0)$ is an integer solution of the given equation also, and x; y; z are relatively prime pair wise. Hence, it suffices to show that the given equation has no non-zero integersolutions (x; y; z) with (x; y) = 1.Suppose that (x; y; z) is such a solution, then x; y cannot be divisible by 3 and $x^2 + y^2 \equiv 2 \pmod 3$, a contradiction. Thus, the conclusion is proven.

6 (CHINA/2003) Given that $\dfrac{1260}{a^2+a-6}$ is a positive integer, where a is a positive integer. Find the value of a

Solution:

$a^2 + a - 6 = (a-2)(a+3)$ implies that (a-2) and (a+3) are both factors of 1260, and their difference is 5. Since $1260 = 2^2 \times 3^2 \times 5 \times 7$, where the pairs of two factors with the difference 5 are (1,6),(2,7),(4,9),(7,12) and (9,14). Thus

a –2 = 1,2,4,7,9 i.e. a = 3,4,6,9,11

7 (CHINA/2001) How many number of pairs (x; y) of two integers satisfy the equation$x^2 - y^2 = 12$?

Solution:

The original equation yields (x –y) (x+y) = 12. Since x – y andx + y have the same parity, and

12 = 2 × 6 = (–2) × (–6), so there are four systems of simultaneous equations :

$$\begin{cases} x-y=2, \\ x+y=6, \end{cases} \quad \begin{cases} x-y=6, \\ x+y=2, \end{cases} \quad \begin{cases} x-y=2, \\ x+y=6, \end{cases} \quad \begin{cases} x-y=-6, \\ x+y=-2, \end{cases}$$

From which four solutions are obtained i.e. (4, 2): (4,–2) ; (–4,–2) and (–4,2).

8 (SSSMO(J)/2004) Let x; y; z and w represent four distinct positive integers such that $x^2-y^2=z^2-w^2=81$. Find the value of xz + yw + xw + yz.

Solution:

The given equations give $(x-y)(x+y) = 3^4$ and $(z-w)(z+w) = 3^4$.

Since x –y < x + y and z –w < z + w

If two of the four numbers x –y, x + y, z –w, z + w have equal values, then the other two must be equal also, and it must be the case that x –y = z –w and x + y = z + w, but then it implies that

y –w = x –z = w –y, i.e. y =w, a contradiction. Thus x –y, x + y, z –w, z + w must be four be four distinct values and further they are the four distinct factors of 3^4 with x + y , z + w being the larger two factors and x – y, z –w being the smaller two. Since $3^4 = 3 \times 1 = 3^3 \times 3$, so

$xz + yw + xw + yz = (x+y)(z+w) = 3^4 . 3^3 = 3^7 = 2187$

9 (CHINA/2003) Find the number of non-zero integer solutions (x; y) to the equation $\frac{15}{x^2y}+\frac{3}{xy}-\frac{2}{x}=2$

Solution:

By eliminating the denominators, the given equation becomes

$15 + 3x - 2xy = 2x^2y$

$2x^2y - 3x + 2xy - 15 = 0$,

$(2xy - 3)(x+1) = 12 = 1\times12 = -1 \times -12 = 3.4 = (-3) \times (-4)$

Since $2xy - 3$ is odd, hence there are four possible systems :

$$\begin{cases} 2xy-3 &= 1, \\ x+1 &= 12, \end{cases} \qquad \begin{cases} 2xy-3 &= -1, \\ x+1 &= -12, \end{cases}$$

$$\begin{cases} 2xy-3 &= 3, \\ x+1 &= 4, \end{cases} \qquad \begin{cases} 2xy-3 &= -3, \\ x+1 &= -4, \end{cases}$$

The first second and the fourth system have no integer solutions, the third system has the solution x = 3, y = 1.

Thus there is exactly one integer solution x = 3, y = 1

10 (CHINA/2001) Find the number of positive integer solutions to the equation $\frac{x}{3}+\frac{14}{y}=3$

Solution:

Simplify the equation to the form xy + 42 = 9y, then

$$y=\frac{42}{9-x}$$

y is a positive integer implies that 9 –x is a positive divisor of 42, so

9–x = 1,2,3,6,7, i.e. = 8,7,6,3,2

Correspondingly, y = 42,21, 14, 7, 6.

By checking the give solutions satisfy the original equation, so the answer is 5.

11 (CHINA/2001) Find the number of positive integer solutions of the equation $\frac{2}{x}-\frac{3}{y}=\frac{1}{4}$

Solution:

The given yields $8y - 12x = xy$, so $xy + 12x - 8y - 96 = -96$, i.e.

$(x - 8)(y + 12) = -96$

Since $y + 12 \geq 13$ and $-7 \leq x - 8 < 0$, there are give possible cases to be considered:

$\begin{cases} x-8 & =-1, \\ y+12 & =96, \end{cases}$ $\begin{cases} x-8 & =-2, \\ y+12 & =48, \end{cases}$ $\begin{cases} x-8 & =-3 \\ y+12 & =32 \end{cases}$

$\begin{cases} x-8 & =-4 \\ y+12 & =24, \end{cases}$ $\begin{cases} x-8 & =-6 \\ y+12 & =16, \end{cases}$

From them five positive integer solutions are obtained easily:

(7,84), (6,36), (5,20), (4, 12), (2,4)

Thus the answer is 5

12 (SSSMO/2005) How many ordered pairs of integers (x; y) satisfy the equation $x^2+y^2=2(x+y)+xy$?

Solution:

By rewriting the equation in the form $x^2 - (2+y)x + y^2 - 2y = 0$, and considering it as a quadratic equation in x (y is considered as a constant in its range), then the equation has integer solutions in x , so its Discriminant is a perfect square.

$$\Delta=(2+y)^2-4(y^2-2y)=4+12y-3y^2=16-3(y-2)^2=n^2$$

For some integer n, it follows that $(y-2)^2 \leq \frac{16}{3}$,so

$$-3<-\frac{4}{3}\sqrt{3} \leq y-2<\frac{4}{3}\sqrt{3}<3$$

Thus, y maybe 0,1,2,3,4

When y = 0, then $x^2 - 2x = 0$ so x = 0 or 2

When y = 1 or 3, then $\Delta = 13$ which is not a perfect square

When y = 2, then $x^2 - 4x = 0$, so x = 0 or 4

When y = 4, then $x^2 - 6x + 8 = 0$, so x = 2 or 4

Thus, there are a total of 6 desired pairs

13 (SSSMO/2003) Let p be a positive prime number such that the equation $x^2-px-580p=0$ has two integer solutions. Find the value of p.

Solution:

Suppose that the two integral roots of the given equation are m and n. Then by Viete Theorem,

m + n = p, (i)

mn =–580 p (ii)

Therefore one of m and n is divisible by p. Without loss of generality, we assume that p |m.

Then m = kp for some integer k.

(i) Yields n = (1–k)p. then

(ii) Yields $(k-1)kp^2 = 580p$, i.e. $(k-1)kp = 580 = 4 \times 5 \times 29$.

Thus p = 29

14 (USSR/1962) Prove that the only solution in integers of the equation $x^2 + y^2 + z^2 = 2xyz$ is x = y = z = 0.

Solution:

Since the sum of the squares is to be an even number, it may be reasoned that either all three of the numbers x^2, y^2, z^2 (hence also x,y,z) are even, or one of them is even and two are odd. But in the last event, the sum would be divisible only by 2 and product 2xyz would be divisible by 4. Hence we must conclude that x,y and z ,must all be even : $x = 2x_1$, $y = 2y_1$, $z = 2z_1$. If we substitute these into the given equation and divisible through by 4, we obtain

$$x_1^2 + y_1^2 + z_1^2 = 4x_1y_1z_1$$

As above, this equation implies that x_1, y_1 and z_1 are all even numbers, and so can write $x_1 = 2x_2$, $y_1 = 2y_2$, $z_1 = 2z_2$, which yields the equation

$$x_2^2 + y_2^2 + z_2^2 = 2^3 x_2y_2z_2$$

Which in turn implies that x_2, y_2 and z_2 are all even numbers, also continuation of this process leads to the conclusion that the following set of numbers is all even:

x,y,z

$$x_1 = \frac{x}{2}, y_1 = \frac{y}{2}, z_1 = \frac{z}{2};$$

$$x_2 = \frac{x}{4}, y_3 = \frac{y}{4}, z_2 = \frac{z}{2};$$

$$x_3 = \frac{x}{8}, y_3 = \frac{y}{8}, z_3 = \frac{z}{8};$$

$$x_k = \frac{x}{2^k}, y_k = \frac{y}{2^k}, z_k = \frac{z}{2^k};$$

(The numbers x_k, y_k, z_k satisfy the equation $x_k^2 + y_k^2 + z_k^2 = 2^{k+1} x_k y_k z_k$) .

But this possible only if x = y = z = 0.

15 (CHINA/1993)The number of positive integer solutions (x, y, z) for the system of simultaneous equations $\begin{cases} xy + xz = 255 \\ xy + yz = 31 \end{cases}$ is:

Option:

(a) 3 (b) 2 (c) 1 (d) 0

Answer: (b)

Solution:

The second equation leads to $y = 1, x + z = 31$ at once.

By substituting them into the first equation it follows that

$x(1+31 - x) = 225$

$x^2 - 32x + 255 = 0,$

$(x-15)(x-17) = 0$

$\therefore x_1 = 15, x_2 = 17$

Thus the solutions are $x = 15, y = 1, z = 16$ and $x = 17, y = 1, z = 14$.The answer is (B)